REST APIs for Beginners



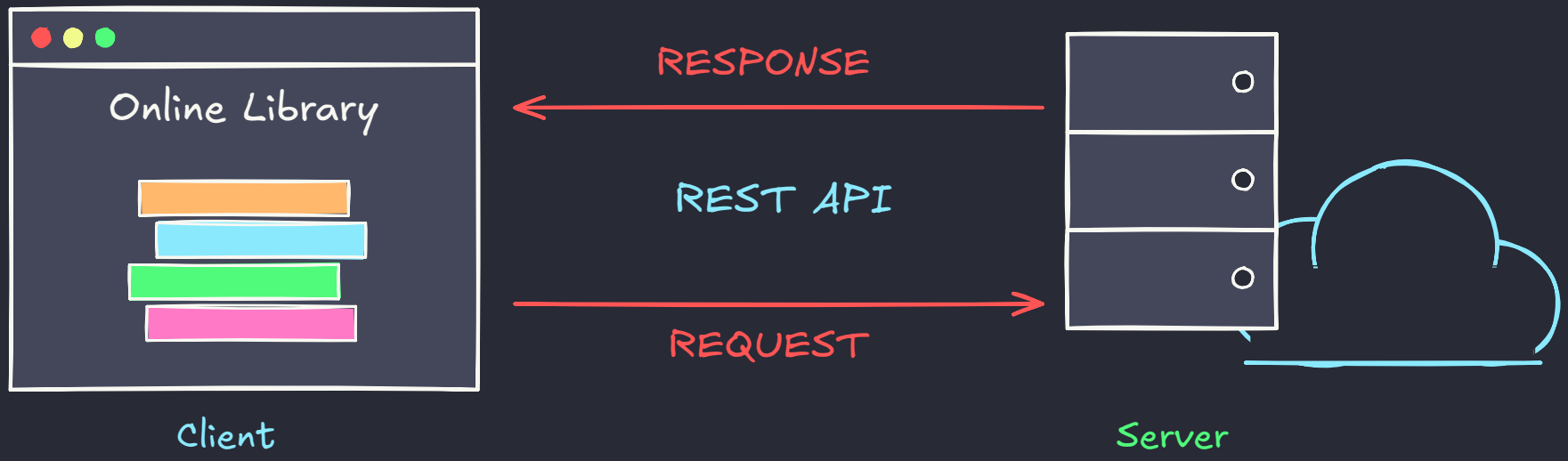
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What is an API?

- **API** = Application Programming Interface
- A set of rules for how one program asks another for something
- You only need to know **what to ask** and **what comes back**
  - not how the other program works inside
- A **web API** does this over the internet

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What is a REST API?



- **REST**: shared rules for how a **client** asks and a **server** answers

“ *Diagram adapted from Rayhan's Tech Realm (2024) — see References* ”

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Watch: REST APIs Explained (in 5 minutes)

- Short walkthrough of REST using an online library: reading, adding, updating and deleting books
- REST APIs Explained (in 5 minutes) — Rayhan's Tech Realm

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Resources and Endpoints

- A **resource** is a thing the API manages — here, **books**
- Each resource has an address, called an **endpoint**

Endpoint	Means
`/books`	all books in the library
`/books/123`	the one book with id 123

- Endpoints are **nouns**; the action comes from the method (next slide)

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CRUD: Four Things You Can Do

Action (CRUD)	HTTP method	Library example
Create	`POST`	Add a new book
Read	`GET`	Look up a book
Update	`PUT`	Change how many copies are available
Delete	`DELETE`	Remove a book

- Same endpoint + different method = different action

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Inside a Request

- **Header** — extra details: data format, who you are (API key)
- **Operation** — the method: `GET`, `POST`, `PUT`, `DELETE`
- **Endpoint** — which resource: `/books/123`
- **Params / body** — data you send, usually as **JSON**

The **response** comes back with a **status code** and, usually, a **JSON body**

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Read a Book — GET



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Add a Book — POST

POST https://api.library.com/books



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Update a Book — PUT

PUT https://api.library.com/books/123



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Remove a Book — DELETE

DELETE https://api.library.com/books/123



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Status Codes: Did It Work?

Code	Meaning	Library example
`200 OK`	Success	Book found
`201 Created`	New resource made	Book added
`204 No Content`	Success, empty reply	Book deleted
`400 Bad Request`	Your data was wrong	Missing title
`401 Unauthorized`	No valid API key	Token missing
`404 Not Found`	No such resource	`/books/9999`
`500 Server Error`	Server broke	Bug on their side

> *Rule of thumb:* ***2xx = worked, 4xx = your mistake, 5xx = their mistake***

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Try It: Call a Real REST API in Python

```python
import requests

url = "https://jsonplaceholder.typicode.com/posts/1"

r = requests.get(url)                          # Read
print(r.status_code, r.json()["title"])

r = requests.put(url, json={"title": "Hi"})    # Update
print(r.status_code, r.json())
```

- **JSONPlaceholder** is a free practice API: no key, nothing really changes
- Swap `get` / `put` for `post` / `delete` to try the other two

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References and Credits

Topic	Source
Diagrams and library example (credit)	Rayhan's Tech Realm. (2024). *REST APIs Explained (in 5 minutes)*. YouTube. Diagrams in this deck are redrawn and adapted from this video.
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HTTP methods	Mozilla. (2026). HTTP request methods — MDN Web Docs.
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Recap

- A **REST API** lets a client ask a server for **resources** by URL
- **Endpoints** are nouns; **methods** are verbs: `GET`, `POST`, `PUT`, `DELETE`
- Each request has a **header**, **operation**, **endpoint** and **body**
- Each response has a **status code** and usually a **JSON body**

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